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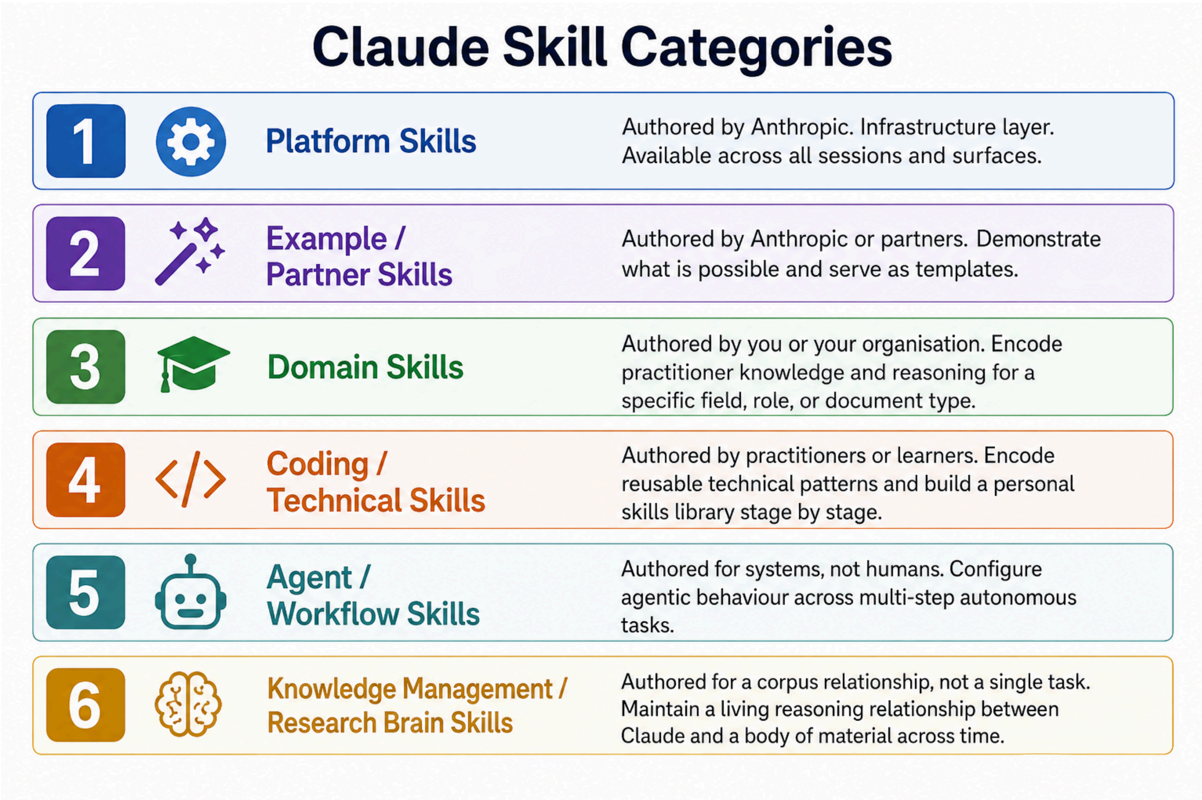
Taxonomy of Claude Skill Categories



A consolidated reference combining all categories, design dimensions, and where they apply — drawn from Erdos Research conversations and practice

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HOW TO READ THIS DOCUMENT

This document defines six content categories of Claude Skills and two design dimensions that cut across all of them. Categories 1–2 are authored by Anthropic. Categories 3–6 are authored by practitioners and organisations. The two dimensions — visibility and invocation surface — are properties every skill carries regardless of category.

The final section provides the one-line summary table for quick reference.

1 Platform Skills

Authored by Anthropic. Infrastructure layer. Available across all sessions and surfaces.

> *These are the foundation layer. Other skill categories call on them rather than duplicate them. A domain skill that produces a report calls on docx; a data analysis skill calls on xlsx.*

Skill	Fires when
`docx`	Creating, reading, or editing Word documents
`pdf`	Creating, filling, or manipulating PDFs
`pdf-reading`	Extracting and inspecting content from PDFs
`pptx`	Creating or reading PowerPoint presentations
`xlsx`	Creating or editing spreadsheets
`file-reading`	Routing uploaded files to the correct reading method by type
`frontend-design`	Building web UIs, React components, HTML/CSS layouts
`product-self-knowledge`	Answering questions about Anthropic's own products accurately

2 Example / Partner Skills

Authored by Anthropic or partners. Demonstrate what is possible and serve as templates for authoring your own.

> *The skill-creator skill is the meta-tool in this category — it is the one you use to build, test, and improve all other skills.*

Sub-type	Examples
Creative tools	algorithmic-art, canvas-design, theme-factory, slack-gif-creator
Builder tools	skill-creator, mcp-builder, web-artifacts-builder
Learning tools	learn
Communication tools	internal-comms, brand-guidelines, doc-coauthoring
Financial tools	financial-calculator
Consumer agents	grocery-shopping, meal-delivery, event-planning, hire-help
Form / workflow agents	file-expenses, file-form, benepass-reimbursement
Service agents	call-to-book, cancel-unsubscribe, return-refund, prescription-refill

3 Domain Skills

Authored by you or your organisation. Encode practitioner knowledge and reasoning for a specific field, role, or document type.

Sub-type	What it encodes	Erdos examples
Methodology skills	A specific analytical or process framework	EAIRM risk assessment, Anti-Pilot evaluation
Role skills	How a professional in a field reasons and decides	Quantum hardware engineer, AI threat analyst
Document / house style skills	Structure and register for a document type	Erdos position paper, Oxford programme brief
Pedagogy skills	A teaching approach, scaffold, or classroom framework	Source analysis framework, PEEL paragraph structure
Persona skills	A named expert with specified depth, stance, and platform tier	The eleven OpenAI / Claude expert personas
Intersection skills	Reasoning that spans two or more fields simultaneously	Post-quantum cryptography (Quantum × Cybersecurity)

4 Coding / Technical Skills

Authored by practitioners or learners. Encode reusable technical patterns. Designed to accumulate stage by stage into a personal skills library.

> *Each stage produces a SKILL.md file the learner keeps. By stage 6 they hold a personal library built from their own domain tasks — the second-brain architecture applied to coding literacy.*

Stage	Skill type	What it encodes
1	Literacy skills	How to read and interpret code
2	Vocabulary skills	How to name and discuss code precisely
3	Prompt-as-spec skills	How to direct Claude to write code for a task
4	Workflow skills	Chained prompt sequences for repeatable tasks
5	Diagnostic skills	How to recover when things break
6	Meta skills	How to build and maintain a personal skills library

5 Agent / Workflow Skills

Authored for systems, not humans. Configure agentic behaviour across multi-step autonomous tasks.

> *Relevant to: Erdos Agent Design Studio, the eleven expert personas document, OpenAI and Claude platform agent work, enterprise AI deployment methodology.*

Sub-type	What it encodes
Persona / identity skills	A named agent identity with specified reasoning style, depth, and platform tier
Task orchestration skills	When to delegate, what to pass forward, when to stop or escalate
Guardrail skills	What the agent refuses, flags, or routes to a human judgement
Integration skills	How the agent calls external tools, APIs, or MCP servers

6 Knowledge Management / Research Brain Skills

Authored for a corpus relationship, not a single task. Maintain a living reasoning relationship between Claude and a body of material across time.

> *The structural distinction from all other categories: this skill never produces a finished output — it maintains and develops a relationship between Claude and a corpus. The primary artefacts are a question log (the researcher's evolving agenda) and living position documents (Claude's current best answer on a given question, not a frozen summary).*

Verb	What it does
ingest	Integrate a new source into the corpus with structured metadata
ask	Retrieve from the corpus at claim level, not document level
position	Generate and continuously revise a living position document on a research question
synthesise	Produce cross-corpus summaries on demand across multiple sources
diff	Compare what the corpus said at an earlier point versus its current state on a question
curate	Flag weak, outdated, or contradicted material across the corpus for review

Relevant to: Erdos AI Researcher Programme, Surrogate Thesis knowledge work, EAIRM framework development, any sustained research engagement spanning 50 or more sources.

D-A Design Dimension A — Visibility and Security

Every skill has a visibility stance — a decision about what logic lives inside the skill file versus behind a service boundary. This is not a category of content but a property every skill carries, and getting it wrong has practical consequences.

Design stance	When to use it	Implication
Fully inspectable	Educational skills, house style, methodology — transparency is the point	Full logic in the SKILL.md. Anyone reading it sees exactly what Claude is asked to do.
Thin interface only	Skills where competitive logic or proprietary methodology must be protected	The skill is an interface only; the actual logic lives behind a backend service the skill calls via MCP or API.
Audit-trail skills	Enterprise governance and compliance — the skill is the written record of intended behaviour	The skill must be auditable; its contents are the accountability document.

> *For Erdos educational skills — position papers, persona scaffolds, pedagogy skills — full inspectability is the right stance and is central to the AI literacy argument: students reading the skill is the lesson. For any commercial deployment where the methodology has competitive value, the thin-interface model applies.*

D-B Design Dimension B — Invocation Surface and Trigger Mode

The same skill artefact behaves differently depending on which surface invokes it. Designing for the right surface is an explicit authoring decision.

Surfaces and invocation mechanisms

Surface	Invocation mechanism
Claude.ai	Pre-built (automatic) or custom-uploaded via Settings
API — Messages	skill_id specified in the API call, pre-built or custom
Claude Code — personal	~/.claude/skills/ — auto-discovered on the user's machine
Claude Code — project	.claude/skills/ inside a repo — shared via version control with the team
Claude Code — plugin	Installed from a marketplace; namespaced to avoid collision across plugins
Claude Code — enterprise	Organisation-pushed via managed settings; overrides personal and project
Agent SDK	Filesystem artefact; opt-in via setting_sources; must exist before the agent runs

Trigger modes

Trigger mode cuts across all surfaces above. Every skill is either model-invoked or user-invoked, and the authoring decisions differ for each.

Trigger mode	How it works	Authoring implication
Model-invoked	Claude reads the skill's description and autonomously decides to load it based on context	Description must be precise enough to fire reliably without over-triggering. The dominant mode on all surfaces.
User-invoked	Explicit triggering: slash command in Claude Code, skill_id in the API, or a direct prompt instruction	Description can be narrower. Suitable for specialist skills the user knows to call by name.

Quick Reference — Full Taxonomy

The table below consolidates the full taxonomy for at-a-glance reference.

#	Category	Core question it answers
1	Platform skills	What file formats and environments does Claude need instructions to handle?
2	Example / partner skills	What has Anthropic demonstrated is possible?
3	Domain skills	What does a practitioner in this field know and how do they reason?
4	Coding / technical skills	What technical patterns does this learner or team use repeatedly?
5	Agent / workflow skills	How should this autonomous system behave across a multi-step task?
6	Knowledge management skills	How should Claude reason with this corpus across time?
D-A	Visibility dimension	What logic can live in the skill versus behind a service boundary?
D-B	Invocation dimension	Where does this skill deploy and how is it triggered?

Overlap note

Categories are not mutually exclusive. A single skill for an Oxford course module is simultaneously a pedagogy skill (category 3), a document skill (category 3), and potentially a role skill if it scaffolds students to reason like a practitioner. The skill-creator skill (category 2) is the tool used to build all others. Platform skills (category 1) are the infrastructure every other category sits on top of. The two dimensions cut across all six categories and must be decided for every skill regardless of which content category it belongs to.

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AI education, applied research, and enterprise deployment methodology.